

PRODUCT & OPERATIONS GUIDE

KisanDost AI

Product Overview Guide

A Plain-Language Guide to Features, User Journeys
and Market Positioning for Non-Technical Stakeholders

✓ FARMER-FIRST, OFFLINE-RESILIENT DESIGN

DOCUMENT CLASS

Product Overview & Operations
Manual

AUDIENCE

Cooperative Managers, District
Officers

RELEASE TARGET

v2.4 Enterprise RC

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1. Project Vision & Mission

KisanDost AI is an all-in-one digital assistant designed for smallholder cattle and dairy farmers. It converts a standard smartphone into a 24/7 pocket veterinarian — allowing farmers to detect animal diseases early, optimize feed recipes, maintain digital health passports, and sell dairy products directly to consumers.

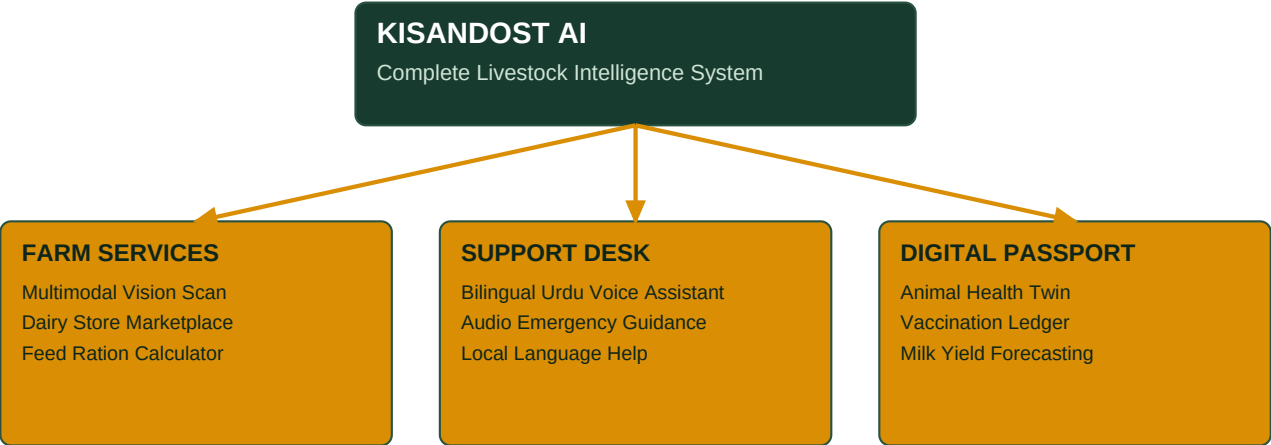


Figure 1. KisanDost AI organizes every capability around three pillars: hands-on farm services, a bilingual support desk, and a portable digital health passport for every animal.

● In one sentence: KisanDost AI puts a pocket veterinarian, a marketplace, and a government-recognized health record in the hands of every smallholder farmer — even when the internet is down.

2. Core Real-World Problems Solved

1. Severe Veterinary Shortage

In agricultural districts, a single certified veterinarian is responsible for up to 15,000 animals. Over 70% of disease outbreaks are diagnosed days too late.

→ **KisanDost AI provides instant disease identification within seconds.**

2. Language & Literacy Barriers

Over 68% of smallholder farmers cannot read English drug leaflets or dosage instructions.

→ **KisanDost AI speaks out loud in localized Urdu and Punjabi.**

3. Unreliable Rural Internet

Cellular connections frequently drop in rural fields, breaking standard cloud-only apps.

→ **KisanDost AI remains fully functional offline, providing local diagnostic aid without freezing.**

4. Middleman Exploitation (Mandi Arthis)

Farmers are often forced to sell high-yield cattle below market value with no proof of health history.

→ **Digital Health Passports generate QR codes proving vaccination history and animal health value.**

3. Detailed Feature Breakdown

Multimodal AI Disease Vision Scanner

Farmers take a photo of an animal showing symptoms — skin lesions, eye discharge, or foot blisters. The AI analyzes the photo and returns:

- Estimated disease identification, with a confidence score
- Immediate emergency level: Low, Moderate, or High
- First-aid and home remedies, explained in Urdu
- Precautions to take before a professional veterinarian arrives

Bilingual Urdu & Punjabi Voice Assistant

Designed for low-literacy users, letting farmers ask questions by voice. The system replies out loud, answering queries on cattle fever, feed recipes, and vaccination schedules.

Digital Health Twin & Milk Yield Forecasting

Tracks body weight, lactation stages, and ambient heat index to predict upcoming milk production curves and detect early risks of mastitis.

Direct Mandi & Dairy Marketplace

An integrated store where farmers list milk, butter, ghee, and verified livestock directly for buyers, eliminating middleman commissions.

4. Dual-Portal User Structure

Farmer Mobile Application

Organized into a streamlined 4-tab layout for ease of use on any smartphone:

Tab	What It Does
Farm Services	Access AI camera diagnostics, feed calculators, and the marketplace.
Support Desk	Interact with the Urdu AI voice assistant for real-time guidance.
User Profile	Manage registered animals, health passports, and vaccination records.
Settings	Toggle between Urdu, Punjabi, and English language options.

Enterprise Administrative Portal

Designed for district veterinary officers and commissioners:

- **Outbreak Radar** — live map tracking regional disease outbreaks and active quarantines.
- **Grievance Redressal** — review and resolve farmer support tickets.
- **Subsidy Management** — verify government dairy certification and subsidy requests.

5. Global & Local Market Positioning

	Pakistan Today	International Standard
Diagnostics	Static SMS helplines, basic classifieds	Real-time AI image analysis
Language Support	English-only prescriptions	Native voice support (KisanDost AI)
Offline Resilience	Fails during network blackouts	Offline-first resilience (KisanDost AI)
Hardware Cost	N/A	\$100+ IoT collar sensors (CattleEye, etc.)

Existing options in Pakistan are mostly static SMS helplines or basic classified ads. KisanDost AI introduces real-time AI image analysis, native voice support, and offline resilience in one product.

Western solutions such as CattleEye rely on high-cost \$100+ IoT hardware collar sensors. KisanDost AI delivers comparable diagnostic insight using only a standard smartphone camera and open-source models.

- **Positioning in one line: the first Pakistani platform to combine AI vision diagnostics, native voice support, and full offline resilience — without requiring any extra hardware.**